Network Configuration Practical

Required Devices

Router: 2911
Switch: 2960
PCs: PC0 and PC1

Step 1: Connect the Devices

1. Connect the router to the switch using a copper straight-through cable.
2. Connect PC0 and PC1 to the switch with copper straight-through cables.

Step 2: Configure IP Addresses

PC0 — IP address: 192.168.10.10; Default gateway: 192.168.10.1
PC1 — IP address: 192.168.20.10; Default gateway: 192.168.20.1

Step 3: Configure the Switch

```
enable
configure terminal

vlan 10
 name HR
exit

vlan 20
 name Cyber
exit

interface fastEthernet 1/1
 switchport mode access
 switchport access vlan 10
 no shutdown
exit

interface fastEthernet 2/1
 switchport mode access
 switchport access vlan 20
 no shutdown
exit

interface fastEthernet 0/1
 switchport mode trunk
 no shutdown
exit

end
```

Step 4: Configure the Router

```
enable
configure terminal

interface fastEthernet 0/0
 no shutdown
exit

interface fastEthernet 0/0.10
 encapsulation dot1Q 10
```

```
 ip address 192.168.10.1 255.255.255.0
exit
```

Step 5: Test Network Connectivity

On PC0, run:

```
ping 192.168.20.10
```

Observed result: Host Unreachable

On PC1, run:

```
ping 192.168.10.10
```

Observed result: Request timed out